



Ain Shams University – Statistical Analysis Course

Project Title: Does sleep duration have a significant relationship with students' academic performance (GPA)?

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Project Summary:

This project investigates whether the number of sleep hours per night affects university students' GPA. Data were collected from many students, including average sleep duration and GPA. Exploratory analysis using histograms, boxplots, and scatterplots described the distributions and visualized the relationship.

Statistical methods such as correlation and linear regression were applied to test the significance of the relationship. The results provide evidence-based conclusions on whether sleep duration predicts academic performance

Data Description:

The data for this project were collected from a sample of many students. The two variables analyzed are **Sleep Hours (X)** and **GPA (Y)**, both of which are quantitative. Sleep Hours represent the average number of hours a student sleeps per night, while GPA measures academic performance on a 0–4 scale. This dataset is suitable for statistical analysis because both variables allow for meaningful numerical and graphical exploration. The chosen variables help investigate whether sleep duration is associated with academic achievement.

Exploratory Data Analysis

Sleeping Hours (X-variable):

Using R code, the following descriptive statistics were obtained for Sleeping Hours.

Sleep Hours per Day

- **n:** 2000
- **Mean:** 7.50 hours
On average, students sleep **7.5 hours per day**, which is close to the recommended amount for young adults.
- **Median:** 7.50
Half of the students sleep **7.5 hours or less**, and half sleep **7.5 hours or more**.
Since the median equals the mean, sleep duration is likely **symmetrically distributed**.
- **Min–Max:** 5.0 to 10.0
The shortest reported sleep duration is **5 hours**, showing the lowest sleepers still get a moderate amount of sleep.
The highest sleep duration recorded is **10 hours**, indicating some students sleep significantly longer.
- **Range:** 5
 $\text{Range} = \text{Max} - \text{Min} = 10 - 5 = \mathbf{5 \text{ hours}}$
This means the total spread of sleep durations is fairly wide, from light sleepers to long sleepers.
- **Q1:** 6.20
25% of students sleep **6.2 hours or less**.
This marks the lower end of the typical sleep range.
- **Q3:** 8.80
75% of students sleep **8.8 hours or less**.
This shows that the top quarter of students sleep nearly 9 hours.
- **IQR:** 2.6

Statistical Analysis

$$\text{IQR} = Q3 - Q1 = 8.8 - 6.2 = \mathbf{2.6 \text{ hours}}$$

The middle 50% of students sleep between **6.2 and 8.8 hours**.

This indicates **moderate variation** within the central portion of the data.

- **Std. Dev: 1.46**

A standard deviation of **1.46 hours** means that students' sleep duration varies moderately around the mean.

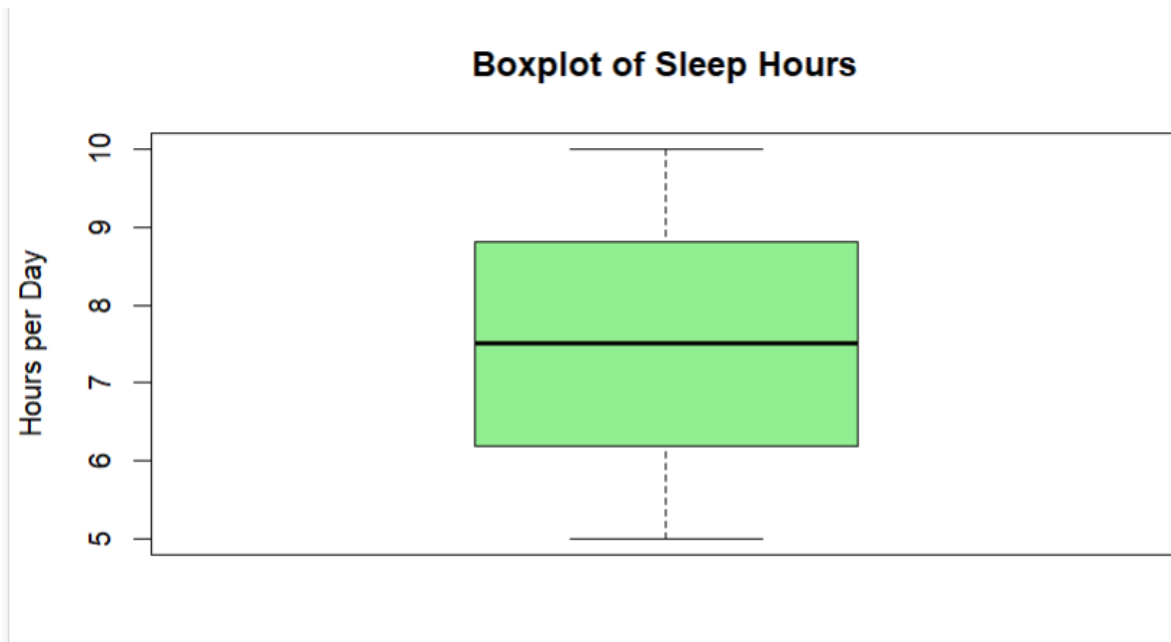
While many sleep between 6 and 9 hours, there are still noticeable differences across individuals.

Distribution Shape:

Sleep is fairly symmetric with slight clustering toward 7–9 hours.

Sleeping hours Graphs:

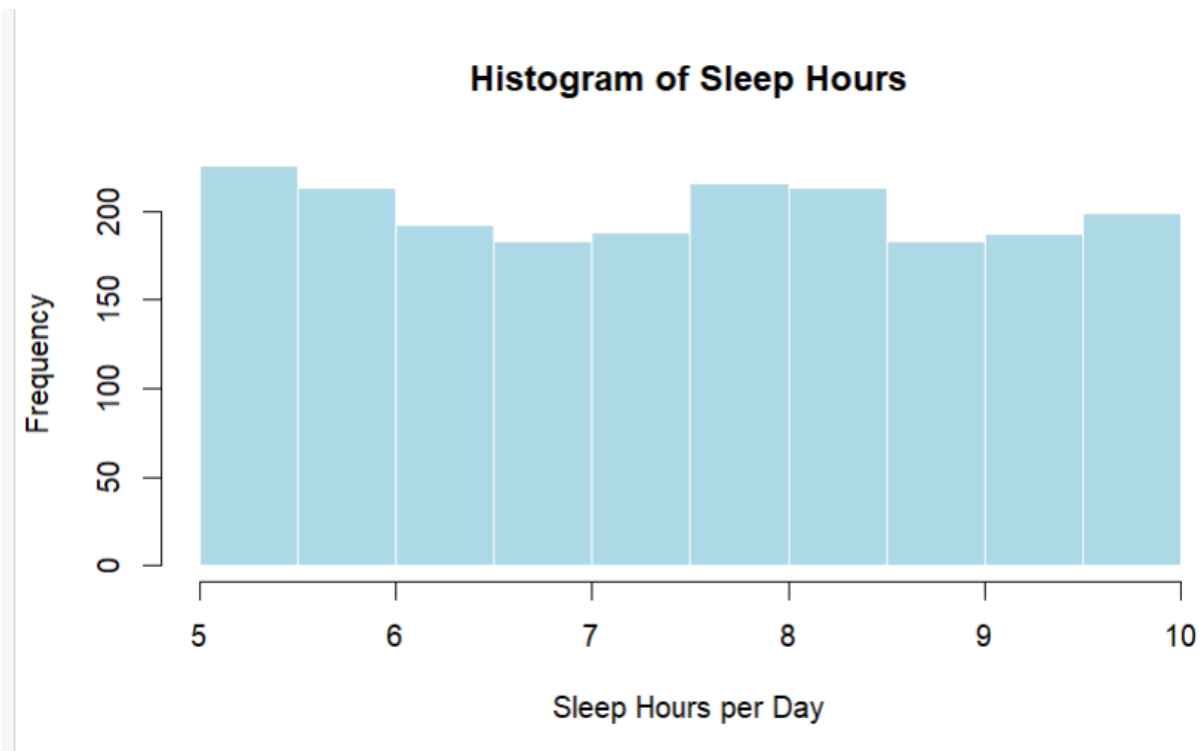
Boxplot of Sleeping Hours



The boxplot shows that most students' sleep durations fall between **6.2 hours (Q1)** and **8.8 hours (Q3)**. The **median of 7.5 hours** lies near the center of the box, indicating a **symmetric distribution** with no strong skewness. The whiskers extend from **5.0 hours** (minimum) to **10.0 hours** (maximum), capturing the full range of reported sleep values.

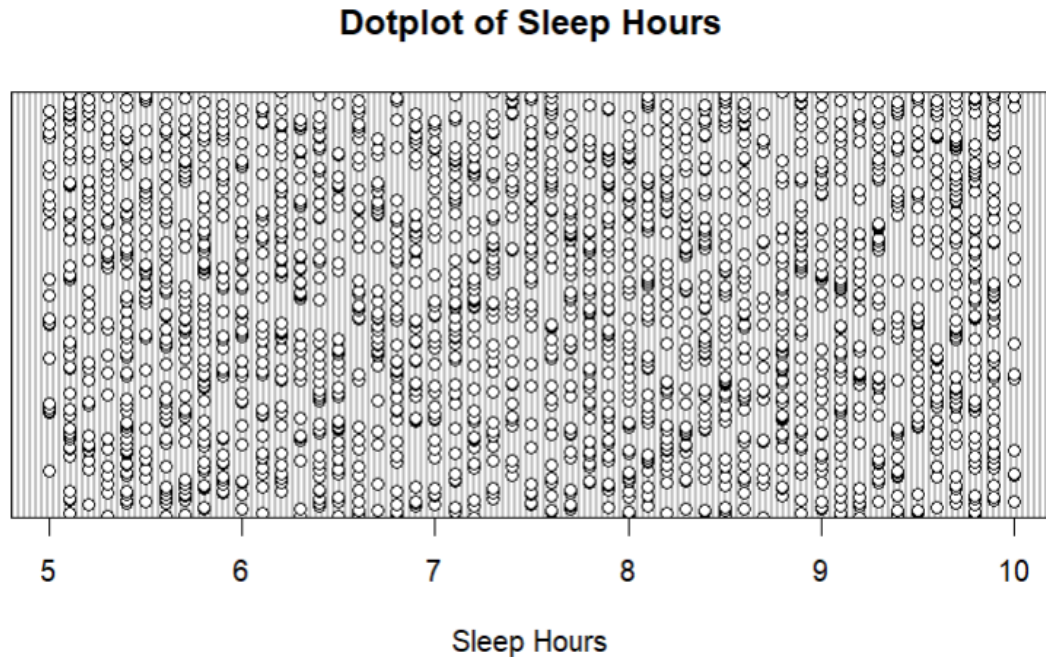
The **IQR of 2.6 hours** shows a moderate concentration of sleep durations around the middle 50% of the data. The whiskers do not show any individual points outside their limits, meaning **no outliers are present**. Overall, the boxplot suggests that students generally sleep between **6 and 9 hours**, with **moderate variation** and a well-balanced distribution around the median.

Histogram of Sleeping Hours



The histogram shows that students' sleep hours are fairly evenly spread between 5 and 10 hours, with most values falling between 6 and 9 hours. The distribution looks balanced with no clear skew, which matches the mean and median both being 7.5 hours. There are small peaks around 6–6.5 and 8–8.5 hours, but no dominant mode, indicating moderate variation in sleep habits.

Dotplot of Sleeping Hours



The dotplot shows that individual sleep-hour values are spread continuously across the full range from 5 to 10 hours, with no noticeable gaps. A large concentration of points appears between 6 and 9 hours, which matches the histogram and reflects where most students fall. The dots are fairly evenly distributed with no isolated extreme points, visually confirming that the dataset contains no outliers. The moderate density across the center also aligns with the IQR of 2.6 hours and the standard deviation of 1.46, indicating a moderate level of variation in sleep duration.

GPA (Y-variable):

Using R code, the following descriptive statistics were obtained for GPA.

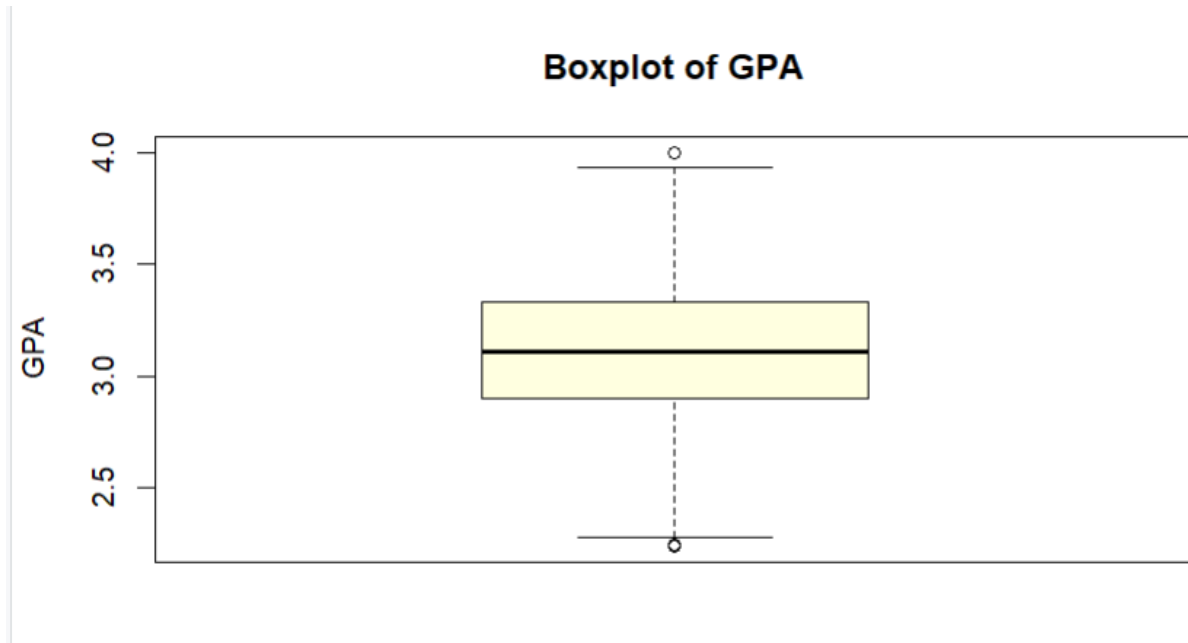
- **n:** 2000
- **Mean:** 3.116
The average GPA of all students is **3.116**, showing that overall academic performance tends to be slightly above average (on a 4.0 scale).
- **Median:** 3.11
Half of the students have a GPA **below 3.11**, and half have a GPA **above 3.11**.
Since the median is very close to the mean, the distribution is likely **symmetrical**, not skewed.
- **Min–Max:** 2.24 to 4.00
The lowest GPA in the dataset is **2.24**, showing that even the weakest students are still within a moderate academic range.
The highest GPA is **4.00**, meaning at least one student achieved a **perfect GPA**.
- **Range:** 1.76
 $\text{Range} = \text{Max} - \text{Min} = 4.00 - 2.24 = 1.76$
This shows the total spread of grades is **not very large**, indicating students' GPAs are relatively close to each other.
- **Q1:** 2.90
25% of students have a GPA of **2.90 or lower**.
This marks the lower boundary of the middle 50% of the data.
- **Q3:** 3.33
75% of students have a GPA of **3.33 or lower**.
This is the upper boundary of the middle 50%.
- **IQR:** 0.43
 $\text{IQR} = \text{Q3} - \text{Q1} = 3.33 - 2.90 = 0.43$
The middle 50% of students have GPAs within a very narrow range, showing **low variation**.
- **Std. Dev:** 0.30
A standard deviation of **0.30** means that most GPAs are close to the mean.
There is **little fluctuation** in students' academic performance.

Distribution Shape:

The mean (3.116) is almost the same as the median (3.11), which suggests a roughly symmetric distribution. The small difference hints at maybe a very slight skew (if any), but given how close these two values are, the distribution is likely approximately symmetric overall.

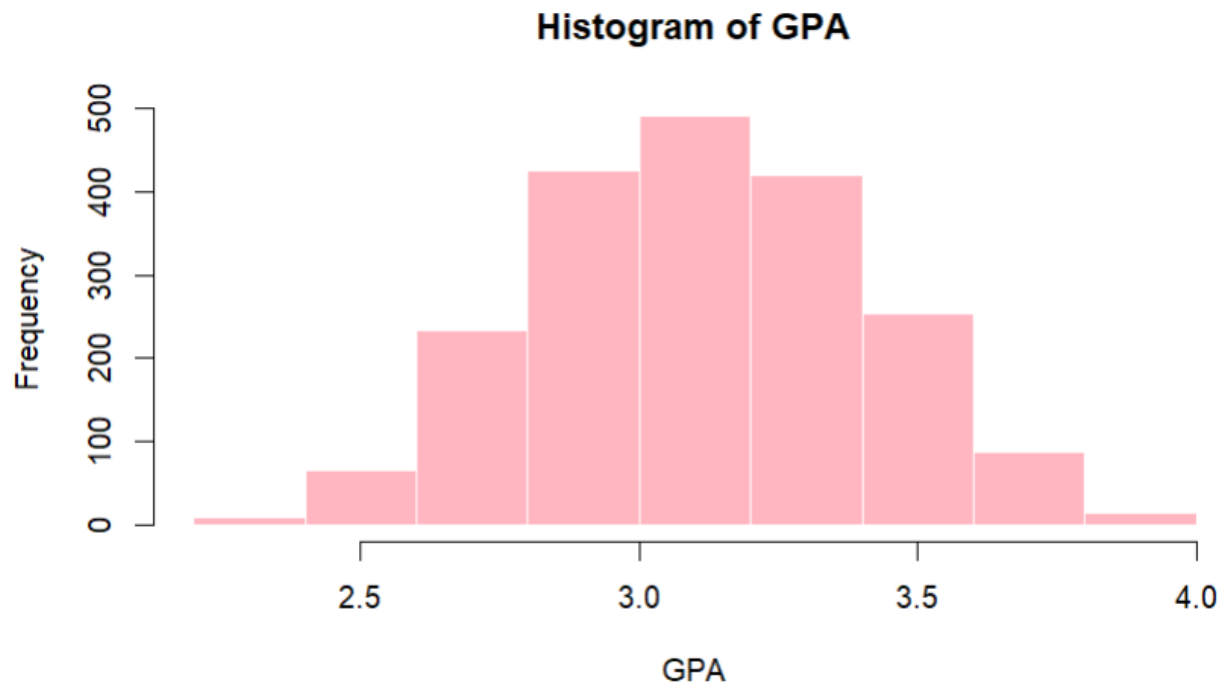
GPA Graphs:

Boxplot:

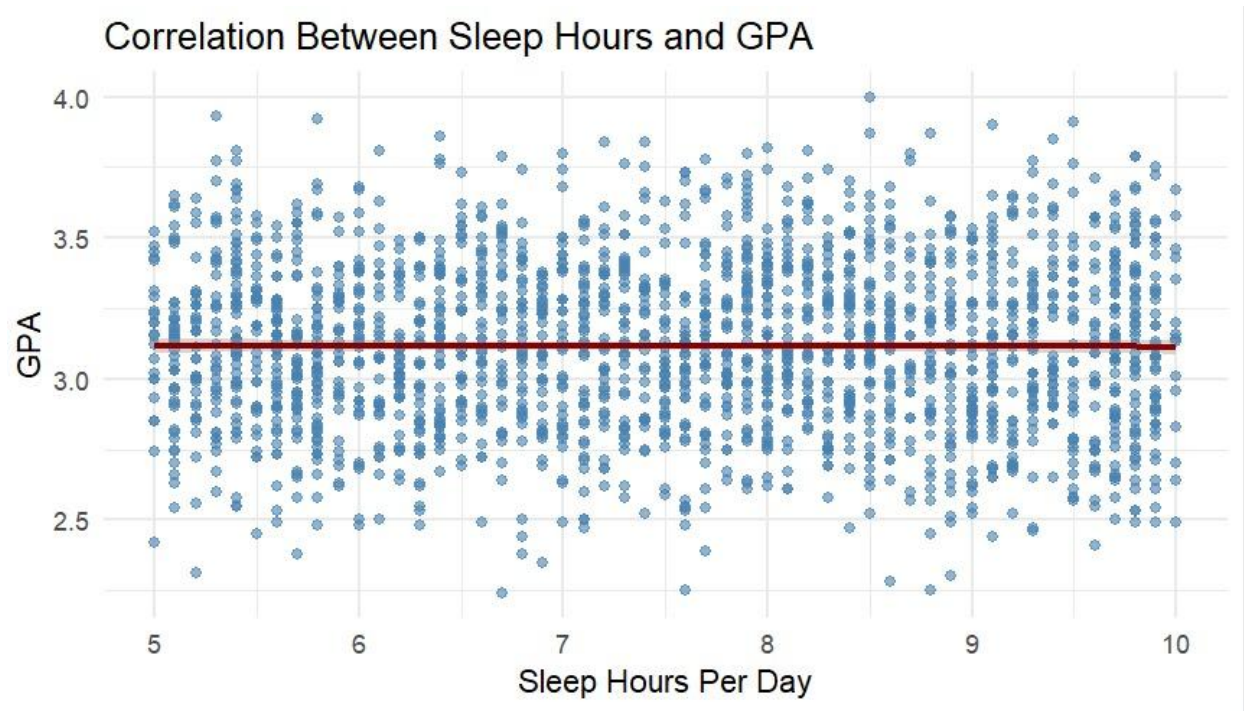


The median GPA is about **3.11**, and the middle 50% of students fall between **2.90 (Q1)** and **3.33 (Q3)**, showing that most GPAs are fairly close together. The whiskers extend from about **2.24** to **4.00**, indicating the overall spread of typical GPAs. A few points outside the whiskers are **outliers**. The median is centered in the box and the whiskers are balanced, so the GPA distribution appears **approximately symmetric** with no strong skew.

Histogram:



The histogram shows that most students have a GPA between **2.8 and 3.4**, with the highest frequency around **3.0–3.2**. The bars form a roughly **bell-shaped** pattern, indicating an **approximately normal distribution**. There is no strong skew, but the distribution slightly tapers off toward higher GPAs (closer to 4.0). Overall, GPAs are concentrated around the average and decrease gradually toward the extremes.



The scatterplot shows **no strong relationship** between sleep hours and GPA. The points are widely scattered across all sleep levels, meaning students with **both low and high GPAs appear at every sleep range**. There is no clear upward or downward trend, so sleep hours do **not strongly predict GPA** in this dataset.

Regression:

Pearson's $r = -0.0043$

- According to the **r-rule**, values close to 0 indicate **no linear relationship**.
- Since -0.0043 is extremely close to zero, the relationship between sleep hours and GPA is **very weak and not meaningful**.
- The negative sign indicates a tiny inverse trend, but it is **too small to matter**.
 r is almost 0 $\rightarrow$ No correlation.

$R^2 = 0.000$ means the model explains **0% of the variance** in GPA.

Sleep hours do **not** help predict academic performance in this dataset.

Conclusion:

To sum up, based on the analysis of the 2000 observations, GPA values show very little variation compared to sleep hours, and the statistical results clearly indicate that there is **no meaningful relationship** between the two variables.

The correlation coefficient of $r = -0.0043$ is extremely close to zero, showing **no linear association** between sleep duration and GPA. This means that changes in sleep hours do **not** correspond to consistent changes in academic performance.

The simple linear regression analysis supports this conclusion.

The slope coefficient ($\beta_1 = -0.0009$) is almost zero, and the p-value (**0.848**) indicates that the relationship is **not statistically significant**.

Additionally, the model's $R^2 = 0.000$ shows that sleep hours explain **none** of the variation in GPA.

Therefore, the answer to the research question is:

No, GPA is not affected by sleep duration in this dataset.

Although sleep is important for overall health, the results here show that, for this particular sample, sleep hours do not play a measurable role in explaining students' academic performance.